Unit Test – Science

Class X · Max. Marks: 40 · Time: 1 hr 30 min

General instructions: (i) All questions are compulsory. (ii) Q15 has an internal choice; attempt only one. (iii) Write question numbers clearly in the margin. (iv) Show all steps in numericals and write SI units.

Section A — Multiple choice (1 mark each)

Q1. Which gas is released during photosynthesis? [1]

(a) Carbon dioxide (b) Oxygen (c) Nitrogen (d) Hydrogen

Q2. The pH of a neutral solution at 25 °C is: [1]

(a) 5 (b) 14 (c) 7 (d) 1

Q3. The SI unit of electrical resistance is: [1]

(a) volt (b) ampere (c) ohm (d) watt

Q4. Which of the following is a decomposition reaction? [1]

(a) $2H_2 + O_2 \rightarrow 2H_2O$
(b) $CaCO_3 \rightarrow CaO + CO_2$
(c) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
(d) $NaOH + HCl \rightarrow NaCl + H_2O$

Q5. The focal length of a concave mirror whose radius of curvature is 30 cm is: [1]

(a) 15 cm (b) 30 cm (c) 60 cm (d) 7.5 cm

Q6. **Assertion (A):** Platinum is used for making jewellery. [1]
Reason (R): Platinum is a highly reactive metal.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Section B — Objective (2 marks each)

Q7. Fill in the blanks: [2]

(i) The energy currency of the cell is _____.
(ii) Rusting of iron is an example of _____ reaction.

Q8. State True or False: [2]

(i) Arteries carry blood away from the heart.
(ii) Bile is secreted by the pancreas.

Q9. Match Column I with Column II: [2]

Column I	Column II
A. Stomata	1. Absorption of digested food
B. Nephron	2. Transmission of nerve impulses
C. Villi	3. Exchange of gases
D. Neuron	4. Filtration of blood

Section C — Short answer (3 marks each)

Q10. (a) Balance the equation: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ [2] [3]

(b) Name the gas liberated in this reaction. [1]

Q11. A resistor of 20 Ω is connected to a 12 V battery. Calculate (i) the current flowing through it and (ii) the power consumed by it. [3]

Q12. Write any three differences between aerobic and anaerobic respiration. [3]

Q13. An object is placed 20 cm in front of a convex lens of focal length 10 cm. Find the position of the image and state its nature. [3]

Q14. (a) What is the role of saliva in digestion? [2] [3]
(b) Name the enzyme present in saliva. [1]

Q15. Explain why the sky appears blue on a clear day. [3]
OR
Explain why stars twinkle but planets do not.

Section D — Long answer (5 marks each)

Q16. (a) Draw a labelled ray diagram to show the image formed by a concave mirror when the object is placed beyond the centre of curvature C. State the position and nature of the image. [3] [5]
(b) Write two uses of concave mirrors. [2]

Q17. (a) What is ozone? Why is depletion of the ozone layer a cause of concern? [2] [5]
(b) Explain the 3Rs to save the environment, with one example of each. [3]

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